

# PIT Embedding Report Aug 21

## PIT-1B Embedding Extraction: Hidden State Normalization

### Summary

While extracting embeddings from PIT-1B for earnings call transcripts, we observed extremely large values in certain hidden dimensions (dim 1531 reaching magnitudes of 10,000+). After investigation, we found this is **not a model deficiency** but an **extraction error**: the model applies `F.rms_norm` to the hidden state before the LM head, and we were extracting the hidden state before this normalization step. Applying `F.rms_norm` to the extracted hidden states produces well-behaved embeddings across all 11 checkpoints (2014-2024).

### Root Cause

PIT is based on the modded-nanogpt architecture, which applies the final normalization as a bare functional call rather than a named module:

```
# PIT's forward method (modeling_pit.py):
for block in self.transformer["h"]:
    x = block(x, ...)
    x = F.rms_norm(x, (x.size(-1),))    # ← functional call, not a
named module
logits = self.lm_head(x)
```

Standard HuggingFace models (GPT-2, LLaMA) implement this as a named module (`self.norm` or `self.ln_f`) and return post-norm hidden states via `output_hidden_states=True`. PIT's implementation does not expose this because:

1. `F.rms_norm` is a bare function call with no learnable parameters, so it does not appear in `model.named_modules()` or `model.state_dict()`
2. `output_hidden_states=True` is not implemented in the custom `PITForCausalLM` (returns `None`)
3. The only way to discover the norm is to read the `forward()` source code

This led us (and likely other researchers using hooks or `output_hidden_states`) to extract the pre-normalization hidden state, producing the large magnitude values.

## Verification Across All Checkpoints

We tested the same 200 earnings call transcripts on all 11 PIT-1B checkpoints (2014-2024), extracting hidden states both with and without `F.rms_norm` applied after the last transformer block.

### With `F.rms_norm` applied (correct extraction)

| Checkpoint | Std  | Max  | Dims > 10 | L2 Norm |
|------------|------|------|-----------|---------|
| 201412     | 0.90 | 32.2 | 2         | 32.7    |
| 201511     | 0.90 | 35.0 | 2         | 34.1    |
| 201612     | 0.90 | 36.1 | 2         | 35.2    |
| 201712     | 0.90 | 36.7 | 1         | 35.9    |
| 201812     | 0.90 | 36.9 | 1         | 36.3    |
| 201912     | 0.90 | 36.8 | 2         | 36.3    |
| 202012     | 0.93 | 36.5 | 2         | 36.4    |
| 202112     | 0.93 | 35.4 | 2         | 36.3    |
| 202212     | 0.93 | 32.6 | 4         | 36.0    |
| 202312     | 0.89 | 27.8 | 5         | 35.4    |
| 202412     | 0.89 | 24.2 | 5         | 35.1    |

Stable across all checkpoints: std ~0.90, L2 norm ~35, at most 5 dimensions exceed abs 10. These are well-behaved embeddings suitable for downstream use.

## Without `F.rms_norm` (incorrect extraction)

| Checkpoint | Std | Max    | Dims > 10 | L2 Norm | Top Dim Mean |
|------------|-----|--------|-----------|---------|--------------|
| 201412     | 255 | 2,314  | 1,536     | 1,803   | 1,610        |
| 201511     | 255 | 4,160  | 1,536     | 3,090   | 2,929        |
| 201612     | 255 | 6,308  | 1,536     | 4,805   | 4,664        |
| 201712     | 255 | 8,924  | 1,536     | 6,975   | 6,825        |
| 201812     | 255 | 10,868 | 1,536     | 8,860   | 8,688        |
| 201912     | 255 | 12,797 | 1,536     | 10,211  | 9,980        |
| 202012     | 291 | 13,687 | 1,536     | 11,368  | 11,025       |
| 202112     | 299 | 13,549 | 1,536     | 11,667  | 11,048       |
| 202212     | 293 | 12,107 | 1,536     | 11,413  | 10,077       |
| 202312     | 268 | 9,469  | 1,536     | 10,600  | 7,907        |
| 202412     | 255 | 7,479  | 1,536     | 9,981   | 5,989        |

The magnitudes grow from 2014 to ~2020 (peaking around 202012-202112) then decrease. All 1,536 dimensions exceed abs 10. These values are entirely driven by the missing normalization step.

## Why This Affects Researchers

The modded-nanogpt architecture (which PIT is based on) was designed for training speed, not for the HuggingFace ecosystem. The implementation gap:

| Feature                           | Standard HuggingFace                                    | PIT (modded-nanogpt)                                 |
|-----------------------------------|---------------------------------------------------------|------------------------------------------------------|
| Final norm                        | <code>self.norm = RMSNorm(...)</code><br>(named module) | <code>F.rms_norm(x, ...)</code><br>(functional call) |
| <code>output_hidden_states</code> | Returns post-norm states                                | Not implemented<br>(returns None)                    |
| Hook on final norm                | Works ( <code>model.norm</code> )                       | Impossible (no module to hook)                       |

| Feature                            | Standard HuggingFace | PIT (modded-nanogpt)            |
|------------------------------------|----------------------|---------------------------------|
| <code>model.named_modules()</code> | Includes norm        | Does not include any final norm |

Any researcher using standard extraction methods (hooks on the last block, `output_hidden_states`, or calling the backbone directly) will get pre-normalization hidden states and observe the large magnitude values.

## Correct Extraction Method

```
import torch
import torch.nn.functional as F

# Hook on last transformer block
captured = {}
def hook_fn(module, input, output):
    captured["h"] = output[0] if isinstance(output, tuple) else output

hook = model.transformer.h[-1].register_forward_hook(hook_fn)
model(input_ids=input_ids, attention_mask=attention_mask)
hook.remove()

# Apply the same rms_norm as the model's forward method
hidden = captured["h"]
hidden = F.rms_norm(hidden, (hidden.size(-1),)) # ← critical step

# Mean pool
mask = attention_mask.unsqueeze(-1).float()
embedding = (hidden * mask).sum(1) / mask.sum(1).clamp(min=1e-9)
```

## Comparison with DatedGPT

With the correct extraction (including `F.rms_norm`), PIT-1B produces embeddings comparable to DatedGPT:

| Metric | PIT-1B (with norm) | DatedGPT |
|--------|--------------------|----------|
| Std    | 0.90               | 0.80     |
| Max    | ~36                | ~24      |

| <b>Metric</b> | <b>PIT-1B (with norm)</b> | <b>DatedGPT</b> |
|---------------|---------------------------|-----------------|
| L2 Norm       | ~36                       | ~36             |
| Dims > 10     | 1-5                       | 4               |

The previous report (Aug 20) incorrectly identified an architectural issue. The model is architecturally sound. The issue was entirely in the extraction method.